

Taewon Yoo

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RESEARCH INTERESTS

Human-Computer Interaction (HCI), Computer-Supported Cooperative Work (CSCW), Social Computing, Data Visualization, Responsible, Sustainable AI

EDUCATION

Yonsei University

M.S. Student, Computer Science and Engineering

- Advisor: Prof. Dae Hyun Kim

Seoul, Korea

Mar. 2025 - Feb. 2027 (Expected)

Kyung Hee University

B.S., Software Convergence (Data Science)

Yongin, Korea

Aug. 2024

WORK EXPERIENCE

Research Intern, University of Seoul

Human-Centered Artificial Intelligence Lab (HCAIL)

- Developed a visual analytics tool supporting ML developers to consider carbon emissions in development
- Developed a smartphone application for dietary assessment for older adults
- Advisor: Prof. Hyunggu Jung

Seoul, Korea

Jun. 2020 - Jun. 2024

Software Developer, Penta Security Systems (alternative military service)

Data Processing Team

- Conducted data analysis of mobility app services to enhance usability and increase user engagement
- Conducted data analysis of EV, delving into battery status, sensor performance, and motor functionality
- Developed a real-time application to calculate taxi fares using geospatial data

Seoul, Korea

Aug. 2021 - Jul. 2023

Research Intern, KIST Europe Forschungsgesellschaft mbH

Smart Convergence Group

- Developed and evaluated a generative adversarial network to solve data-imbalance problem
- Advisor: Prof. Sungho Suh

Saarbrücken, Germany

Aug. 2019 - Jan. 2020

PUBLICATIONS

International Journal Papers

1. S. Na, S. Park, J. Jun, H. Jeong, **T. Yoo**, M. Chun, H. Lee, H. Lee, H. Jung. “SUV: Students’ Understanding Visualizer to Support Instructors in Synchronous Online Lectures”. *IEEE Access*, August 2023. DOI: 10.1109/ACCESS.2023.3303030
2. M. Chun, H. Jeong, H. Lee, **T. Yoo**, H. Jung. “Development of Korean Food Image Classification Model Using Public Food Image Dataset and Deep Learning Methods”. *IEEE Access*, December 2022. DOI: 10.1109/ACCESS.2022.3227796

International Conference Papers

1. S. Lee*, **T. Yoo***, E. Park. “Inside and Outside the Virtual Study Room: An Exploratory Study of Student Practices and Perceptions”. In *Publication of the 2025 European Conference on Computer-Supported Cooperative Work and Social Computing (ECSCW’25), EUSSET*. (To Appear) (*Equally contributed)
2. **T. Yoo**, N. Augustat, B. Hogedorn. “Bridging EU Climate Policy and AI Development: Toward Design Implications for Collaborative Sustainable AI Communities”. In *Companion Publication of the 2025 European Conference on Computer-Supported Cooperative Work and Social Computing (ECSCW’25 Companion), EUSSET*. (To Appear)
3. **T. Yoo**, H. Lee, S. Oh, H. Kwon, H. Jung. “Visualizing the Carbon Intensity of Machine Learning Inference for Image Analysis on TensorFlow Hub”. In *Companion Publication of the 2023 Conference on Computer-Supported Cooperative Work and Social Computing (CSCW’23 Companion)*. ACM, Minneapolis, MN, USA, October 2023. DOI: 10.1145/3584931.3606959
4. **T. Yoo**, M. Chun, Y. Bae, S. Kwon, H. Jung. “Exploring the Community of Model Publishers on TensorFlow Hub”. In *Companion Publication of the 2022 Conference on Computer Supported Cooperative Work and Social Computing (CSCW’22 Companion)*. ACM, Virtual, October 2022. DOI: 10.1145/3500868.3559477

5. **T. Yoo**, H. Jeong, D. Lee, H. Jung. "LectYS: A System for Summarizing Lecture Videos on YouTube". In *Proceedings of the 26th International Conference on Intelligent User Interfaces (IUI'21 Companion)*. ACM, Virtual, April 2021. DOI: 10.1145/3397482.3450722
6. **T. Yoo**, H. Jung. "Category Classification of Educational Videos on YouTube through Machine Learning Approaches using Video Titles". In *Proceedings of the US-Korea Conference on Science, Technology and Entrepreneurship (UKC)*, Chicago, IL, USA, August 2019.

Domestic Conference Papers

1. **T. Yoo**, H. Jung. "Exploring the Community of Sustainable AI Developers on Hugging Face". In *Proceedings of the HCI Korea 2024 Conference*, Hongcheon, Republic of Korea, January 2024. **(BEST PAPER AWARDS)**
2. **T. Yoo**, D.Lee. "Video Summarization with Unpaired Dataset and Adversarial Network". In *Proceedings of Korea Software Congress (KSC) 2023*, Busan, Republic of Korea, December 2023.
3. **T. Yoo**, H. Jung. "Category Classification of Educational Videos on YouTube through Machine Learning Approaches". In *Proceedings of Korea Computer Congress (KCC) 2019*, Jeju, Republic of Korea, June 2019.
4. **T. Yoo**, H. Jung. "Needs Analysis Based on Vlogging Career". In *Proceedings of the HCI Korea 2019 Conference*, Jeju, Republic of Korea, February 2019.

ACADEMIC SERVICES

Reviewer

CSCW 2024 Poster, CHI 2025 LBW

Student Volunteer

CHI('24, '25), CSCW('21-'23), RecSys('21, '22), FAccT('22), IUI('21), ASSETS('21), DIS('21)

PRESENTATIONS

Sustaining Scalable Sustainability Workshop, CHI'24

Visualizing the Carbon Intensity of Machine Learning Inference for Image Analysis on TensorFlow Hub

EXTRACURRICULAR ACTIVITIES

Teaching Assistant

CAS4102 AI Capstone Design, Yonsei University

Spring 2025

SWCON104 Web and Python Programming, Kyung Hee University

Spring 2019, 2020, 2021

GEE1920 Freshmen Seminar, Kyung Hee University

Spring 2018, 2019

GEE1952 Major Exploring Program, Kyung Hee University

Fall 2018

Vice President of Student Association

Department of Software Convergence, Kyung Hee University

2018

HONORS AND AWARDS

Best Paper Awards

HCI Korea 2024

2024

Excellence Prize

Hyundai AutoEver (Barrier Free App Development Contest)

2022

The National Scholarship for Science and Engineering

Full tuition fee for 4 semesters, Korea Student Aid Foundation

2019

The National Scholarship for Excellence in Software

National Program of Excellence in Software

2019

Encouragement Awards

Blue-print Ideathon

2019

Humanitas Scholarship

Kyung Hee University

2018, 2019

Bright Society Scholarship

Kyung Hee University

2018